

Place Value, Rounding and Estimating — Paper A

MEDIUM • NON-CALCULATOR • 40 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer every question in the space provided. Show your working — method marks are awarded even when the final answer is wrong.
- No calculator. Every number on this paper is designed to be handled on paper.
- The decimal point does not move. The digits move: one column left for each $\times 10$, one column right for each $\div 10$.
- $\times 0.1$ is the same as $\div 10$, and $\div 0.1$ is the same as $\times 10$. Rewrite the question that way before you calculate.
- Read every rounding instruction twice: decimal places and significant figures give different answers on the same number.
- Write a zero before the decimal point (0.47, never .47), and keep placeholder zeros (38 000, never 38).
- The mark for each question is shown in square brackets on the right.

SECTION 1 – POWERS OF TEN, AND 0.1

1. Complete the table. Each empty box is worth one mark. [6]

Number	$\times 100$	$\div 100$
5.3		
47		
0.6		

Count the zeros to count the columns. The point stays put; the digits travel.

2. Work out:

(a) 8.2×10 [1]

(b) $3.5 \div 100$ [1]

(c) 0.09×1000 [1]

(d) $620 \div 10\,000$

[1]

3. Work out:

(a) 7.4×0.1

[1]

(b) $5 \div 0.1$

[1]

(c) 3.8×0.01

[1]

(d) $6 \div 0.01$

[1]

SECTION 2 – ROUNDING

4. Round 4736 to:

(a) the nearest 10

[1]

(b) the nearest 100

[1]

(c) the nearest 1000

[1]

(d) 1 significant figure

[1]

5. Round 12.3648 to:

(a) 1 decimal place [1]

(b) 2 decimal places [1]

(c) 3 decimal places [1]

(d) 2 significant figures [1]

6. Round each of these to 2 significant figures.

(a) 0.06382 [1]

(b) 5471 [1]

(c) 0.9047 [1]

(d) 38.72 [1]

SECTION 3 – ESTIMATING

7. Estimate 61.4×4.8 by rounding each number to 1 significant figure. Show what you rounded each number to. [2]

- 8.** Estimate $3891 \div 19$ by rounding each number to 1 significant figure. Show what you rounded each number to. [2]

End of paper. Check every answer has a unit or a form the question actually asked for.